

Screenshot 2025-08-14 135923

Polygon rendering is the process of calculating intensity and considerations for a polygon surface.

Scene Description + Illumination Model + Rendering Technique = Image

There are two ways of polygon rendering:

<< 'Render each polygon surface with single intensity, -or

Calculate intensity at each point of the surface using interpolation scheme

In order to produce a realistic image with various kinds of reflection, there are 3 common shading methods which are mainly applied to polygons:

- 1 Constant Intensity shading Method.
2. Gouraud Shading method (Intensity Interpolation) and.
3. Phong Shading Method (Normal Vector Interpolation),

Application of an illumination model to the rendered

formed with polygon surfaces are key techniques for polygon rendering

the surface normal at each visible point and applying the desired illumination model at that

Point is expensive. We can describe more efficient shading models for surfaces defined by

polygons and polygon meshes. Scan line algorithms typically apply a lighting model to obtain

polygon surface rendering in one of two ways. Each polygon can be rendered with a

constant intensity, or the intensity can be obtained at each point of the surface using an interpolation

scheme,